MODERN RITHMOMACHIA

Rules of play

2 PLAYERS · AGES 10 AND UP · 30 TO 45 MINUTES

Rithmomachia, the "battle of numbers", was the learned game of medieval schools. This modern version keeps its spirit: pieces that carry numbers, captures based on relations between numbers, and victory by lining up a progression. Everything is on the board; nothing is left to chance.

Components

An 8 × 8 board and 32 pieces: 16 white, 16 black. Each side has 8 circles, 4 triangles and 4 squares, carrying the numbers **2, 3, 4 and 6**. Each number appears four times per side: two circles, one triangle, one square.

Shape	Pieces per side	Range
Circle	8: two 2s, two 3s, two 4s, two 6s	1 square
Triangle	4: one 2, one 3, one 4, one 6	1 or 2 squares
Square	4: one 2, one 3, one 4, one 6	1, 2 or 3 squares

Setup

Set up the pieces as in diagram 1: squares in the corners, triangles in the middle of the second row, circles 2 and 4 in front.

White: squares 6 a1, 3 b1, 2 g1, 4 h1; circles 3 a2, 6 b2, 6 g2, 3 h2; triangles 4 c2, 2 d2, 3 e2, 6 f2; circles 2 c3, 4 d3, 4 e3, 2 f3.

Black: the mirror image, on rows 8, 7 and 6.

The thick line marks the middle of the board. Rows 5 to 8 are **Black's half**, rows 1 to 4 **White's half**.

White moves first; then the players alternate, one move per turn.

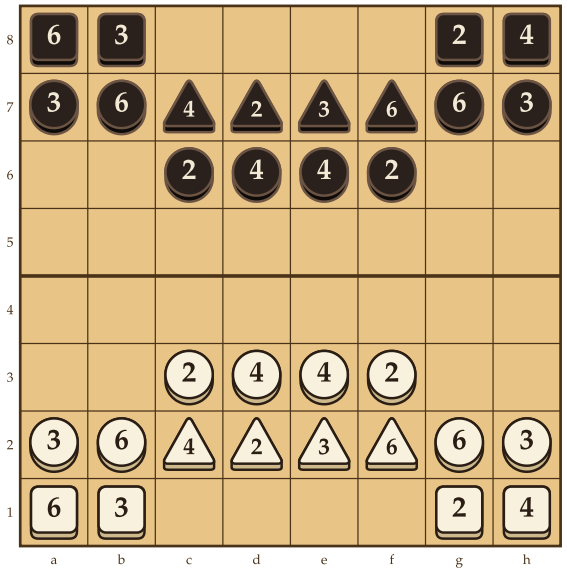


Diagram 1 — The starting position.

Movement

On your turn, move one of your pieces **in a straight line**, in any of the eight directions (row, column, diagonal), up to its range: 1 square for a circle, 1 or 2 for a triangle, 1 to 3 for a square.

A piece never jumps over another piece. It cannot stop on a piece of its own side. It may stop on an enemy piece only to capture it by encounter or by assault (see below).

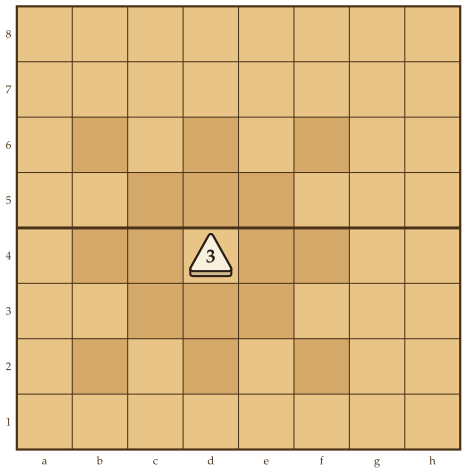


Diagram 2 — The triangle reaches the 16 marked squares.

Object of the game: the triumph

You win at once when, after your move, **three of your pieces touch in a line** (row, column or diagonal) **in the opponent's half**, and their numbers form a **progression**, in either direction:

2-3-4 · 2-4-6 · 2-3-6 · 3-4-6

The first two are arithmetic progressions (the same amount is added each time), the other two harmonic progressions. The middle number is always a 3 or a 4. The shape of the pieces does not matter.

All three pieces must stand in the opponent's half: for White, on rows 5 to 8.

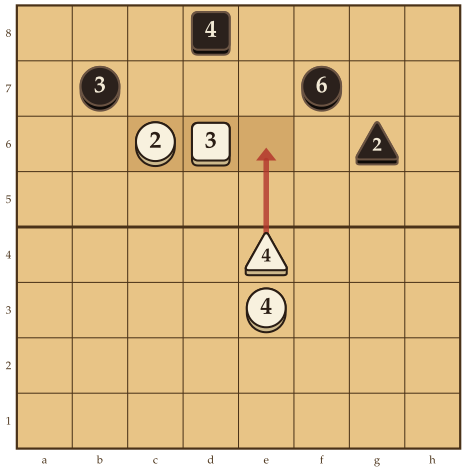


Diagram 3 — The triangle 4 moves up from e4 to e6. Circle 2, square 3 and triangle 4 touch on row 6, in Black's half: 2-3-4, White triumphs.

Captures

There are three ways to capture an enemy piece, based on three relations between numbers: **equality** (encounter), **product** (assault) and **sum or difference** (ambush). A captured piece is removed from the board.

Capturing is never compulsory. A single move can make several captures, for instance an assault followed by an ambush, or two ambushes.

Encounter — equality

A piece captures by landing on an enemy piece **with the same number and a different shape**, within its range, along a clear path.

Two identical pieces, for example two circles 4, can never capture each other.

The encounter defends as much as it attacks: a piece protected by a friendly piece with the same number and a different shape cannot be taken without a recapture.

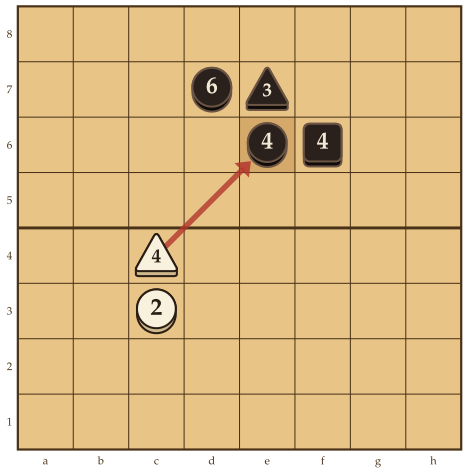


Diagram 4 — The triangle 4 travels two squares and captures the circle 4. The black square 4 can recapture by encounter.

Assault — multiplication and division

A piece captures by landing on an enemy piece **exactly 2 squares** away if that piece is **double or half** its own number, or **exactly 3 squares** away if it is **triple or a third**. The path must be clear.

Circles, which move only one square, can never assault. Only squares reach 3 squares.

Attacker	at 2 squares	at 3 squares
2	a 4	a 6
3	a 6	—
4	a 2	—
6	a 3	a 2

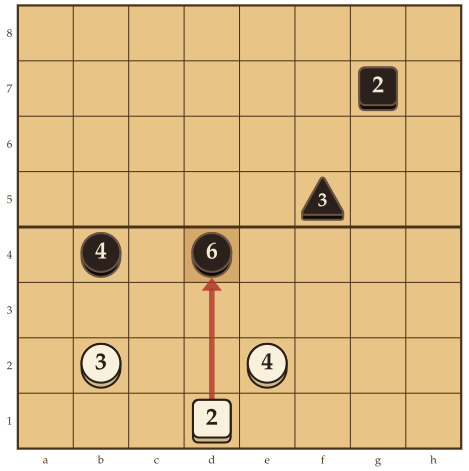


Diagram 5 — The square 2 travels three squares: 2 × 3 = 6, it captures the circle 6.

Ambush — addition and subtraction

After a move, if the piece just moved touches an enemy piece and a friendly piece stands **right behind it**, on the same line, the enemy piece is captured when its number is the **sum or the difference** of the numbers of the two pieces around it. Shapes do not matter.

- **Only the piece just moved springs an ambush.** You may safely move a piece between two enemy pieces.
- One move can spring several ambushes at once, in several directions.

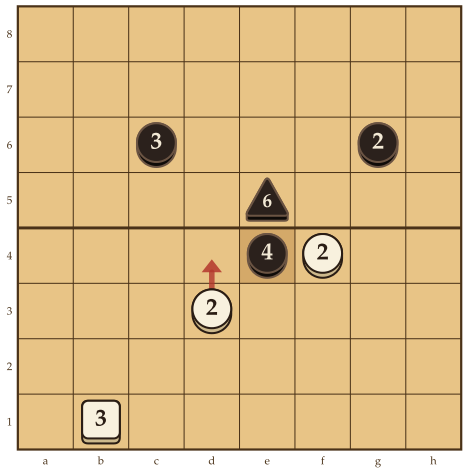


Diagram 6 — The circle 2 moves to d4, against the black circle 4; the white circle 2 on f4 closes the ambush: 2 + 2 = 4, the circle 4 is captured.

End of the game

- **Triumph**: the player who forms a triumph wins.
- **Exhaustion**: a player left with fewer than three different numbers can no longer form a progression and loses.
- **Blocked**: a player who has no legal move loses.
- **Draw**: after 60 consecutive plies without any capture (30 moves by each player), the game is drawn.

Who captures what

Target	Encounter	Assault (2 squares · 3 squares)	Ambush
2	a 2 of another shape	a 4 · a 6	4 and 2, 6 and 4
3	a 3 of another shape	a 6 · —	6 and 3
4	a 4 of another shape	a 2 · —	2 and 2, 6 and 2
6	a 6 of another shape	a 3 · a 2	2 and 4, 3 and 3

Tips for beginners

- **Move forward.** Triumphs are formed in the opponent's half: a piece that stays at home can only defend.
- **Keep all four numbers.** The last piece of a number is worth far more than the others: without it your progressions disappear, and each lost number brings you closer to exhaustion.
- **Watch for triumph threats**: two enemy pieces in line in your half, with a free square beside them, announce a triumph on the next move.
- **Before capturing, ask who recaptures** on the landing square: by encounter, assault or ambush.

Notation. Squares are named from White's side: columns a to h, rows 1 to 8. Shapes are written C (circle), T (triangle), S (square). “C3 a2-b3”: the circle 3 moves from a2 to b3. “S2 g1×g4”: the square 2 captures on g4. “C4 d3-d4 (×e5)”: the move captures the piece on e5 by ambush.

Going further: the *Modern Rithmomachia Player's Manual* (tactics, strategy, openings, endgames, 28 puzzles) and the online game on the website.

A game by Guillaume Enguehard, after medieval rithmomachia · guillaume-enguehard.github.io